

We formulate prompt complexity C as the number of in-context samples. For example, $C = 1$ means we only add one in-context sample. For prompting setup and postprocessing details, we follow the standard implementation in Qwen-2.5-Math (Yang et al., 2024).

Cognac (Chen et al., 2022) is a controlled generation task requiring language model *not* to generate specified banned words provided in the prompt. We use the WordNet subset (Miller, 1995) of Cognac as this is the only released setting in Cognac paper, where the topic is a root node and the constraint is defined as a subtree. We sampled 200 instances using the provided data generation codes in our experiments. To ensure most model generations ended properly in the decoding process, we relax the constraint of maximum decoded tokens T from 60 to 512. We use the same prompt templates following their Github repo.¹¹

Creative Story Generation (Chakrabarty et al., 2024) provides the plots and story continuation from both machine and human. We adopt the provided 11 human-written story plots in the original dataset as the prompt. In this task, we set the maximum token $T = 1024$ to ensure the continued story written by LLM can have a proper ending. We formulate prompt complexity C as providing $C \times 25$ words in the plot.

Random Strings Similar to Bigelow et al. (2024), we sample 200 random strings with length $L \sim U(256, 512)$ from the tokenizer vocabulary as the prompt. Prompt complexity C is formulated by providing $C \times 15$ tokens in the prompt, ensuring each article contains at least 100 tokens.

BBCLatestNews (Li et al., 2024b) is a news collection dataset aims at collecting news that is beyond the time cut for training LLMs. Unlike creative story plots, news articles are typically more structured and organized, although headlines can still be surprising. We select news articles from January to July 2024 to minimize data contamination, as the Llama models have a knowledge cut-off in late 2023. We formulate prompt complexity C as providing $C \times 15$ words in the prompt.

C Entropy Underestimation via Monte Carlo Sampling

To demonstrate the limitations of Monte Carlo (MC) sampling for entropy estimation in long sequences, we conducted an empirical study. We prompted Llama-3-8B-Instruct with 5-shot CoT examples from the MMLU dataset. We then estimated the entropy of its generated responses using a varying number of MC samples: $M \in \{4, 8, 16, 32, 64\}$.

As illustrated in Figure 10, the estimated entropy consistently increases with the number of samples. This trend confirms that MC estimation with a small sample size systematically **underestimates** the true entropy because it fails to capture the long tail of the full probability distribution. While increasing the sample count mitigates this bias, it does so at a significant computational cost. In contrast, Theorem 3.1 allows us to use the negative log-likelihood (NLL) of a single typical sequence for a more efficient and accurate estimation.

D Full BF Output Dynamics Investigation

Here we present full task-wise and model-wise BF output dynamic for Llama-2 in Figure 8 and Llama-3 in Figure 9. We can observe the trends as in § 4.1: ① **The average BF for the base model (≈ 12) is roughly ten times higher than the aligned model (≈ 1.2).** ② **BF would often drop smoothly as more output tokens are generated.**

E Curious Case of Prompt Complexity

Intuitively, greater prompt specificity (larger C) reduces BF by narrowing the model’s output space through more informative context. However, our experimental results reveal task-varied effects. As illustrated in Figure 11 for the Cognac task, greater prompt complexity can *increase* BF—potentially due to the cognitive burden of processing negation or complex linguistic structures. In contrast, for tasks like News

¹¹<https://github.com/princeton-nlp/Cognac/tree/main>

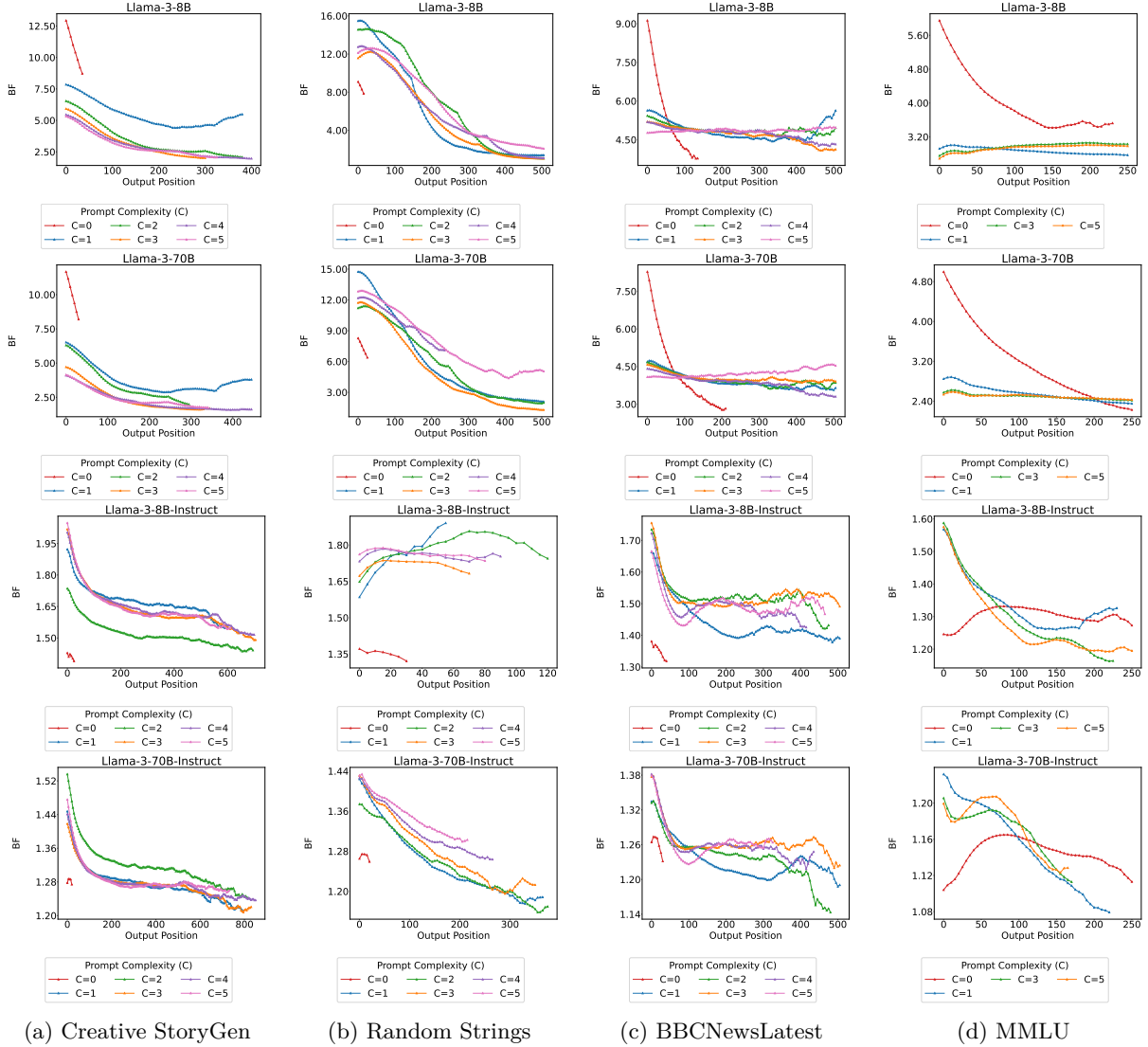


Figure 9: **BF Output Dynamic for Llama-3-families.** For better visualization, we compute the exponential moving averaged values of perplexity with the smoothing factor set as 0.1.

Generation, higher C generally leads to lower BF, consistent with the expected narrowing of output diversity. Comprehensive task-wise BF results are provided in § F.

F Full Task-wise BF Evaluation on Different Prompt Complexity

The full task-wise BF evaluation results over different prompt complexity can be found in Figure 12. Here we can see that prompt complexity modulates BF in highly non-consistent ways across models and tasks, and there are no clear monotonic patterns, contradicting the intuition that with more context given, the model should have more confidence in what to generate.

G Generalization to Additional Tasks and Models

To confirm the generalizability of our findings (§ 4), we extend our experiments to new domains: summarization on XSUM (Narayan et al., 2018), multilingual tasks on AYA (Singh et al., 2024). We formulate prompt

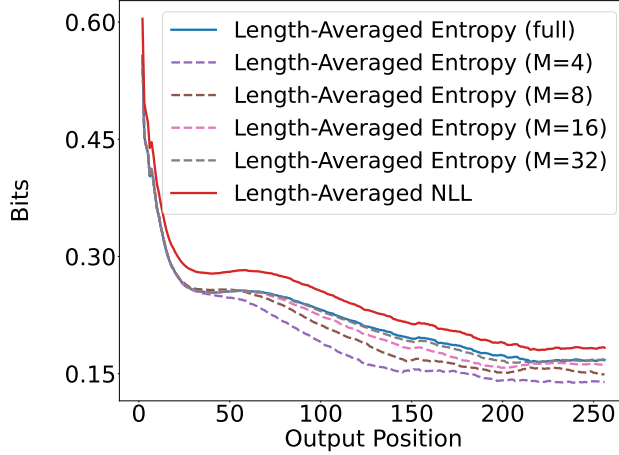


Figure 10: **Monte Carlo (MC) sampling systematically underestimates entropy.** The plot shows that the estimated entropy of sequences from Llama-3-8B-Instruct increases with the number of MC samples (M). A small sample size fails to cover the vast output space, leading to an underestimation of the true entropy. This bias is difficult to eliminate without incurring substantial computational costs.

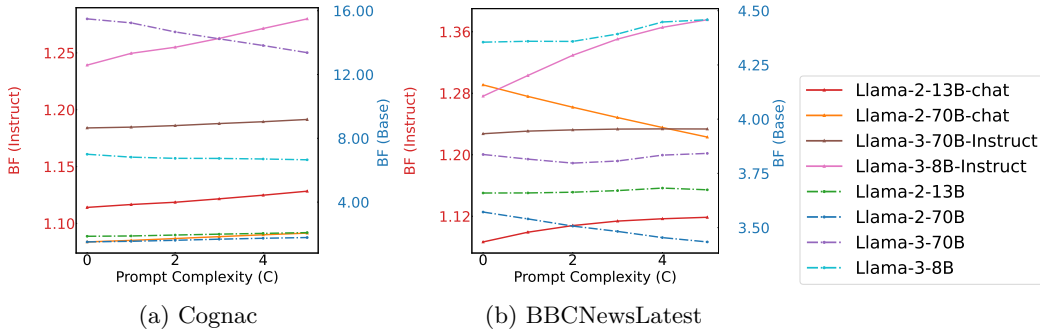


Figure 11: **Task-varied influence of prompt complexity C on BF.** On Cognac, we see BF increases with increased C , while on BBCNewsLatest, increasing C can lead to reduced BF.

complexity C as providing $C \times 25$ words in the prompt. We also verify our findings on a new model, Qwen3-4B (Qwen, 2025).¹² As presented in Figure 13, our core conclusions remain robust across these diverse conditions.

We also analyze OLMo-2 (OLMo et al., 2024) across different alignment stages (Base and DPO) on Creative StoryGen and MMLU, covering both 7B and 13B scales. As shown in Figure 14, OLMo-2 exhibits a similar trend where alignment tuning reduces BF, although the reduction is less pronounced compared to Llama models, suggesting that OLMo-2’s post-training asserts less influence on the generation manifold. Additionally, we provide the BF dynamics for Qwen3-4B on MMLU in Figure 15.

H Proof of LLM Log-Likelihood Convergence

The following proof is a simplified version of the one in (Mudireddy et al., 2024), presented for completeness and to refine its original bounds. For the formal measure-theoretical treatment, we refer readers to the original paper. While a more direct proof using the weak law of large numbers is possible, we use Chebyshev’s inequality to provide a more self-contained and accessible argument.

¹²For the Qwen3 family, we use the Qwen3-4B-Base and Qwen3-4B-Instruct-2507 pair. Other aligned variants can be activated into a reasoning mode, exhibiting behavior distinct from the models in our main study, and were thus excluded for a fair comparison.